EFGPND Conditioning Note

This note updates the earlier σ_n^2 memo with the current picture:

- the original noise-trace pathology was real, and the specialized feature-space formula fixed most of it;
- the remaining hard piece is now the shared mean solve
$$(G + \sigma_n^2 I)\beta = DF^* y,$$
not the old σ_n^2 trace block;
- this is primarily a large-n conditioning problem;
- factoring out $1/\sigma_n^2$ by itself does *not* improve the condition number of that mean system;
- the next algebraic opportunity for the remaining lengthscale problem is to differentiate the *whole approximate objective* in feature space, rather than only rewriting the trace term.

1. Operators

Write the approximate kernel in feature form as
$$\widetilde{K} = \Phi\Phi^*, \qquad \Phi = FD, \tag{1}$$
and define
$$G = \Phi^*\Phi = D(F^*F)D. \tag{2}$$
Then the data-space and feature-space systems are
$$K = \widetilde{K} + \sigma_n^2 I = \Phi\Phi^* + \sigma_n^2 I, \tag{3}$$
$$A_{\text{mean}} = G + \sigma_n^2 I. \tag{4}$$
The shared solve used to build
$$\alpha = K^{-1} y$$
is the feature-space system
$$A_{\text{mean}}\beta = DF^* y, \qquad \alpha = \frac{1}{\sigma_n^2}\left(y - FD\beta\right). \tag{5}$$
This same α enters the quadratic-form part of *every* hypergradient:
$$y^\top K^{-1}\left(\frac{\partial K}{\partial \theta}\right)K^{-1} y = \alpha^\top\left(\frac{\partial K}{\partial \theta}\right)\alpha. \tag{6}$$

2. Why Large n Makes This Bad

The current difficulty is most apparent in large dense datasets such as the PRISM raster because the top eigenvalues of $G = D(F^*F)D$ grow with the sampling density. On a fixed domain with infill sampling, this means they grow roughly with n, up to kernel-dependent constants and the scale set by σ_f^2.

So the mean system has condition number

$$\kappa(A_{\text{mean}}) \approx \frac{\lambda_{\max}(G) + \sigma_n^2}{\sigma_n^2}. \tag{7}$$

This becomes bad for three separate reasons:

- large n increases $\lambda_{\max}(G)$;
- large σ_f^2 increases the scale of G;
- small or moderate σ_n^2 leaves a weak floor on the bottom of the spectrum.

This explains why the problem is much more visible on PRISM than on smaller datasets even when the number of features M is only a few hundred.

3. Why Small Residual Does Not Mean Good Gradient

CG controls the residual

$$r = b - A\hat{x},$$

not the solution error directly. For a linear system $Ax = b$,

$$\frac{\|x - \hat{x}\|}{\|x\|} \lesssim \kappa(A)\,\frac{\|r\|}{\|b\|}. \tag{8}$$

So if $\kappa(A_{\text{mean}})$ is on the order of 10^7 to 10^8, then a relative residual tolerance such as 10^{-4} or even 10^{-5} is a weak guarantee for the quality of $K^{-1}y$.

That is exactly what the fixed-state PRISM checks show: many CG iterations, apparently small residuals, and still a poor approximation to the true feature-space solve.

4. Why the σ_n^2 Fix Worked

The earlier noise-variance problem was different. The old stochastic trace term for σ_n^2 required repeated solves with harsh probe right-hand sides of the form

$$A_{\text{mean}}^{-1}(DF^*z). \tag{9}$$

After the algebraic rewrite, the trace identity becomes

$$\operatorname{tr}(K^{-1}) = \frac{n}{\sigma_n^2} - \frac{1}{\sigma_n^2}\operatorname{tr}(A_{\text{mean}}^{-1}G), \tag{10}$$

and the corresponding Hutchinson estimator uses solves of the form

$$A_{\text{mean}}^{-1}(Gv). \tag{11}$$

The gain did *not* come mainly from pulling out the factor $1/\sigma_n^2$. The real gain came from changing the problem structurally:

- an exact closed-form term n/σ_n^2 was peeled off;
- the remaining solves use Gv, so the right-hand side lives in range(G);
- weak or null directions are suppressed instead of being excited by raw white-noise probes.

That is why the σ_n^2 trace block became dramatically easier.

5. Does Factoring Out $1/\sigma_n^2$ Help the Mean Solve?

Not by itself.

The code already has the scaled operator

$$A_{\text{var}} = I + \frac{1}{\sigma_n^2} G, \tag{12}$$

implemented as

`create_A_var`.

But

$$A_{\text{mean}} = G + \sigma_n^2 I = \sigma_n^2 A_{\text{var}}. \tag{13}$$

Since multiplying a matrix by a positive scalar does not change its condition number,

$$\kappa(A_{\text{mean}}) = \kappa(A_{\text{var}}). \tag{14}$$

So simply factoring out $1/\sigma_n^2$ does *not* cure the mean-solve conditioning problem. It only rescales the operator.

6. Current Empirical Picture on Late PRISM

At a representative late PRISM state

$$\ell = 0.09256, \qquad \sigma_f^2 = 3.878, \qquad \sigma_n^2 = 0.05202, \qquad M = 625, \tag{15}$$

I compared `compute_gradients` against the exact feature-space solve for the *current approximate objective*, using the same Hutchinson probes.

The exact raw/log-parameter gradient was approximately

$$g_{\text{exact}} \approx \begin{bmatrix} 1.40 \times 10^6 \\ -7.19 \times 10^4 \\ -6.36 \times 10^5 \end{bmatrix}. \tag{16}$$

The observed relative raw-gradient errors were:

setting	mean CG iters	trace CG iters	relative raw-gradient error
baseline, `cg_tol` $= 10^{-4}$	753	178	2.86×10^2
preconditioned, `cg_tol` $= 10^{-4}$	166	50	3.21×10^1
preconditioned + warm start, `cg_tol` $= 10^{-4}$	69	50	2.76×10^1
preconditioned, `cg_tol` $= 10^{-5}$	977	194	3.22×10^{-1}
preconditioned + warm start, `cg_tol` $= 10^{-5}$	685	194	2.21×10^{-1}

So:

- preconditioning helps a lot;
- warm starting helps further;
- but late in training, even `cg_tol` $= 10^{-5}$ is still only moderately accurate, not exact.

7. The Next Algebraic Win: Special-Case σ_f^2

The signal variance is also algebraically special. If

$$\widetilde{K} = \sigma_f^2 \widetilde{K}_0, \tag{17}$$

then

$$\frac{\partial K}{\partial \sigma_f^2} = \frac{\widetilde{K}}{\sigma_f^2} = \frac{K - \sigma_n^2 I}{\sigma_f^2}. \tag{18}$$

Therefore

$$\operatorname{tr}\left(K^{-1} \frac{\partial K}{\partial \sigma_f^2}\right) = \frac{1}{\sigma_f^2} \operatorname{tr}\left(I - \sigma_n^2 K^{-1}\right) = \frac{n - \sigma_n^2 \operatorname{tr}(K^{-1})}{\sigma_f^2}, \tag{19}$$

and

$$\alpha^\top \left(\frac{\partial K}{\partial \sigma_f^2}\right) \alpha = \frac{1}{\sigma_f^2} \left(\alpha^\top K \alpha - \sigma_n^2 \alpha^\top \alpha\right) = \frac{y^\top \alpha - \sigma_n^2 \alpha^\top \alpha}{\sigma_f^2}. \tag{20}$$

So the positive-parameter gradient for σ_f^2 is

$$\frac{\partial}{\partial \sigma_f^2} \log p(y \mid \theta) = \frac{1}{2\sigma_f^2} \left(n - \sigma_n^2 \operatorname{tr}(K^{-1}) - y^\top \alpha + \sigma_n^2 \alpha^\top \alpha\right). \tag{21}$$

In raw/log-parameter space, this becomes

$$g_{\log \sigma_f^2} = \sigma_f^2 \frac{\partial}{\partial \sigma_f^2} \log p(y \mid \theta) = \frac{1}{2}\left(n - \sigma_n^2 \operatorname{tr}(K^{-1}) - y^\top \alpha + \sigma_n^2 \alpha^\top \alpha\right). \tag{22}$$

Likewise, the raw/log-parameter noise gradient is

$$g_{\log \sigma_n^2} = \sigma_n^2 \frac{\partial}{\partial \sigma_n^2} \log p(y \mid \theta) = \frac{\sigma_n^2}{2}\left(\operatorname{tr}(K^{-1}) - \alpha^\top \alpha\right). \tag{23}$$

This means that once α and $\operatorname{tr}(K^{-1})$ are available, both σ_f^2 and σ_n^2 can be handled *without* their own stochastic derivative blocks. Only the lengthscale still needs the hard derivative-specific trace machinery.

8. Methodological Implication

The main lesson is not merely "use more CG iterations." The better methodology is:

- exploit algebraic structure for special hyperparameters whenever possible;
- avoid stochastic derivative solves for σ_n^2 and σ_f^2;
- recognize that the remaining hard core is the shared mean solve and the lengthscale derivative, both of which are limited by conditioning rather than by missing algebra.

So the most promising path forward is:

1. keep the specialized σ_n^2 treatment;
2. special-case σ_f^2 as above;
3. then focus numerical work on the mean solve $(G + \sigma_n^2 I)\beta = DF^* y$, where large-$n$ conditioning is now the real bottleneck.

9. Lengthscale: What Failed

After special-casing σ_n^2 and σ_f^2, the remaining hyperparameter-specific difficult block is the lengthscale trace term

$$\frac{1}{2} \operatorname{tr}\left(K^{-1} \partial_\ell K\right).$$

The first idea was the direct analogue of the successful σ_n^2 rewrite. Let

$$C = F^* F, \qquad D = \operatorname{diag}(w), \qquad A = DCD + \sigma_n^2 I, \tag{24}$$

and write

$$\partial_\ell \widetilde{K} = F S_\ell F^*, \qquad S_\ell = \operatorname{diag}(d_\ell), \qquad d_\ell = \frac{\partial (w^2)}{\partial \ell}. \tag{25}$$

Using the Woodbury form

$$K^{-1} = \sigma_n^{-2}\left(I - \Phi A^{-1}\Phi^*\right), \qquad \Phi = FD, \tag{26}$$

one gets the exact identity

$$\operatorname{tr}\Big(K^{-1}\partial_\ell K\Big) = \frac{1}{\sigma_n^2}\operatorname{tr}(CS_\ell) - \frac{1}{\sigma_n^2}\operatorname{tr}\Big(A^{-1}DCS_\ell CD\Big). \tag{27}$$

This identity is correct, but the resulting estimator is not numerically useful in the low-probe regime of interest.

Why It Failed in Practice

The problem is that the full lengthscale derivative is *sign-changing*. So the formula above is an exact constant minus a noisy stochastic correction for an indefinite operator. At $J = 1$, this creates very large cancellation variance.

On a small exact check with real data ($N = 192$, $M = 625$), the true lengthscale trace was about

$$\operatorname{tr}\Big(K^{-1}\partial_\ell K\Big) \approx -3.576 \times 10^3.$$

Over 1000 seeds with $J = 1$:

- the old estimator had mean/std approximately

$$-3.575 \times 10^3 \;\;/\;\; 7.10 \times 10^2;$$

- the full feature-space rewrite had mean/std approximately

$$-3.055 \times 10^3 \;\;/\;\; 8.82 \times 10^4.$$

So the direct rewrite is algebraically right, but statistically terrible at $J = 1$.

The same pattern showed up on the frozen late-PRISM state used above ($\ell = 0.09256$, $\sigma_f^2 = 3.878$, $\sigma_n^2 = 0.05202$, $M = 625$). Over 20 seeds with $J = 1$:

- the old lengthscale trace estimator had sample mean/std approximately

$$-7.38 \times 10^5 \;\;/\;\; 1.17 \times 10^6;$$

- the full rewrite had sample mean/std approximately

$$-1.13 \times 10^9 \;\;/\;\; 1.01 \times 10^{10}.$$

That is why it destabilized training even though the underlying identity was correct.

10. A Better Route: Exact Scale Term + PSD Residual

The more promising route is *not* to rewrite the whole lengthscale derivative at once. Instead, split it into:

- an exact amplitude-like piece proportional to $\widetilde{K}$;
- a residual piece that is positive semidefinite in feature space.

Suppose the lengthscale derivative of the spectral weights can be written as

$$\frac{\partial(w^2)}{\partial \ell} = c_\ell\, w^2 - q_\ell, \qquad q_\ell(\xi) \geq 0 \ \text{ for all } \xi. \tag{28}$$

Equivalently,

$$\partial_\ell \widetilde{K} = c_\ell \widetilde{K} - F Q_\ell F^*, \qquad Q_\ell = \operatorname{diag}(q_\ell). \tag{29}$$

Then the trace term becomes

$$\operatorname{tr}\Big(K^{-1}\partial_\ell K\Big) = c_\ell \operatorname{tr}(K^{-1}\widetilde{K}) - \operatorname{tr}\Big(K^{-1} F Q_\ell F^*\Big) \tag{30}$$

$$= c_\ell \left(n - \sigma_n^2 \operatorname{tr}(K^{-1})\right) - \operatorname{tr}\Big(K^{-1} F Q_\ell F^*\Big). \tag{31}$$

The first term is exact once $\operatorname{tr}(K^{-1})$ is known from the stable σ_n^2 machinery.

For the residual term, the same Woodbury manipulation gives

$$\operatorname{tr}\Big(K^{-1} F Q_\ell F^*\Big) = \frac{1}{\sigma_n^2} \operatorname{tr}(C Q_\ell) - \frac{1}{\sigma_n^2} \operatorname{tr}\Big(A^{-1} D C Q_\ell C D\Big). \tag{32}$$

This has the same basic structure as the successful noise rewrite, but now the operator being estimated is associated with the *positive* residual Q_ℓ, not the full sign-changing derivative.

Why This Is More Plausible

This route avoids the thing that killed the full rewrite: it does not ask one Hutchinson estimator to represent the entire indefinite lengthscale derivative as a single constant-minus-noisy correction.

Instead:

- the exact amplitude-like piece is peeled off deterministically;
- the stochastic part is only the residual PSD operator

$$F Q_\ell F^*;$$

- the feature-space solve still uses the same mean-system matrix

$$A = DCD + \sigma_n^2 I.$$

That is much closer in spirit to the σ_n^2 rewrite, which was stable because it estimated a filtered PSD object rather than a sign-changing one.

11. SE and Matérn Both Fit This Pattern

Squared Exponential

For the squared exponential kernel,

$$S(\xi) = \sigma_f^2 \left((2\pi)\ell^2\right)^{d/2} \exp\left(-2\pi^2\ell^2\|\xi\|^2\right), \tag{33}$$

so

$$\frac{\partial S}{\partial \ell} = S(\xi)\left(\frac{d}{\ell} - 4\pi^2\ell\|\xi\|^2\right). \tag{34}$$

Therefore

$$\frac{\partial(w^2)}{\partial \ell} = \frac{d}{\ell}w^2 - \underbrace{4\pi^2\ell\|\xi\|^2 w^2}_{q_\ell(\xi)\geq 0}. \tag{35}$$

So the SE kernel fits the split exactly with

$$c_\ell = \frac{d}{\ell}, \qquad Q_\ell = \operatorname{diag}\left(4\pi^2\ell\|\xi\|^2 w^2\right). \tag{36}$$

Matérn

For the Matérn family in the parameterization used here,

$$S(\xi) = \sigma_f^2 c_{\nu,d}\, \ell^d \left(2\nu + 4\pi^2\ell^2\|\xi\|^2\right)^{-(\nu+d/2)}, \tag{37}$$

for a positive constant $c_{\nu,d}$ independent of ℓ and ξ. Differentiating gives

$$\frac{\partial S}{\partial \ell} = S(\xi)\left(\frac{d}{\ell} - \frac{8\pi^2(\nu+d/2)\,\ell\|\xi\|^2}{2\nu + 4\pi^2\ell^2\|\xi\|^2}\right). \tag{38}$$

So again

$$\frac{\partial(w^2)}{\partial \ell} = \frac{d}{\ell}w^2 - \underbrace{\frac{8\pi^2(\nu+d/2)\,\ell\|\xi\|^2}{2\nu + 4\pi^2\ell^2\|\xi\|^2}w^2}_{q_\ell(\xi)\geq 0}. \tag{39}$$

Therefore the same exact-plus-residual strategy is not just SE-specific; it extends naturally to Matérn as well.

12. What Computation Would Actually Change

The key term to change is still only

$$\frac{1}{2}\operatorname{tr}\left(K^{-1}\partial_\ell K\right).$$

Under the proposed split, this becomes:

- an exact scale term
$$\frac{c_\ell}{2}\left(n - \sigma_n^2 \operatorname{tr}(K^{-1})\right),$$
- plus a residual trace term built from
$$H_\ell^{\mathrm{res}} = DCQ_\ell CD.$$

The stochastic residual is still a Hutchinson trace estimate in feature space:

$$\operatorname{tr}\left(A^{-1} H_\ell^{\mathrm{res}}\right) \approx \frac{1}{J}\sum_{j=1}^{J} v_j^* A^{-1} H_\ell^{\mathrm{res}} v_j. \tag{40}$$

So there is still:

- a stochastic trace term;
- a CG solve with the same operator
$$A = DCD + \sigma_n^2 I;$$
- one solve per probe.

What changes is the right-hand side construction. The residual right-hand side is

$$b_j^{\mathrm{res}} = H_\ell^{\mathrm{res}} v_j = DCQ_\ell CDv_j, \tag{41}$$

which uses only Toeplitz applications of $C = F^*F$ and diagonal multiplications.

13. Complexity and Expected Benefit

This residual rewrite would *not* fix the shared mean solve

$$(G + \sigma_n^2 I)\beta = DF^* y,$$

which is still the main common bottleneck.

But it would make the lengthscale-specific trace work look like

$$O(JM \log m) + O(k_{\mathrm{cg}} JM \log m), \tag{42}$$

up to moderate constants, while keeping the stochastic part tied to a PSD residual operator rather than the full sign-changing derivative.

So the best current picture is:

- the full direct lengthscale rewrite is mathematically correct but too noisy at low J;
- the natural next attempt is a *partial* rewrite: exact scale term plus PSD residual;
- that route generalizes cleanly to SE and Matérn, but in its naive form it is still not good enough at $J = 1$.

Current Status of the PSD-Residual Split

I tested this split directly against exact ground truth for the approximate EFGP objective, both on small dense real-data subsets and on the full PRISM state with exact $M \times M$ systems.

The result is disappointing: the naive split is still dramatically noisier than the old estimator in the $J = 1$ regime.

On the small SE check ($N = 192$, $M = 625$), the exact lengthscale trace was about

$$-3.576 \times 10^3.$$

Over 1000 seeds with $J = 1$:

- the old estimator had mean/std approximately

$$-3.594 \times 10^3 \ / \ 7.00 \times 10^2;$$

- the PSD-residual split had mean/std approximately

$$-1.506 \times 10^3 \ / \ 1.24 \times 10^5.$$

On the small Matérn check ($N = 192$, $M = 3481$):

- the old estimator had mean/std approximately

$$-1.973 \times 10^3 \ / \ 2.06 \times 10^2;$$

- the PSD-residual split had mean/std approximately

$$-2.464 \times 10^3 \ / \ 4.33 \times 10^4.$$

On the frozen full-PRISM state, the exact lengthscale trace was about

$$-3.628 \times 10^4.$$

Over 20 seeds with $J = 1$:

- the old estimator had mean/std approximately

$$-3.606 \times 10^4 \ / \ 7.65 \times 10^3;$$

- the PSD-residual split had mean/std approximately

$$-4.673 \times 10^8 \ / \ 1.95 \times 10^9.$$

So the practical conclusion is now:

- the full rewrite fails;

- the exact-scale-plus-PSD-residual split is conceptually cleaner, but still fails badly as a low-probe drop-in estimator;
- positive semidefiniteness of the residual alone is not enough to make the resulting Hutchinson estimator usable at $J = 1$.

That means the next improvement for lengthscale probably has to come from a different direction: better probe design, correlated probes, control variates, or a different optimization strategy for the lengthscale parameter itself, rather than another straightforward constant-plus-feature-space-trace rewrite.

14. Whole-Objective Feature-Space Derivative

The next algebraic candidate is to stop splitting the lengthscale gradient into "trace term" and "quadratic term" in data space, and instead differentiate the entire approximate objective directly in feature space.

Write

$$K = \sigma_n^2 I + \Phi\Phi^*, \qquad \Phi = FD, \qquad G = \Phi^*\Phi = DCD, \qquad C = F^*F. \tag{43}$$

Define also

$$A = G + \sigma_n^2 I, \qquad b = DF^*y. \tag{44}$$

Then the approximate negative log marginal likelihood can be written exactly as

$$\mathcal{L}(\theta) = \frac{1}{2}\left[(n-M)\log\sigma_n^2 \;+\; \log\det A \;+\; \frac{1}{\sigma_n^2}\left(y^\top y - b^* A^{-1} b\right)\right] + \text{const.} \tag{45}$$

Now consider any hyperparameter θ that changes only the spectral weights D and not σ_n^2. Then

$$\partial_\theta \mathcal{L} = \frac{1}{2}\left[\operatorname{tr}(A^{-1}A_\theta) - \frac{1}{\sigma_n^2}\left(2\,\Re(b_\theta^*\beta) - \beta^* A_\theta \beta\right)\right], \qquad \beta = A^{-1}b. \tag{46}$$

Here

$$A_\theta = D_\theta C D + D C D_\theta, \qquad b_\theta = D_\theta F^* y. \tag{47}$$

It is better numerically to avoid differentiating w directly. Since $w^2 = s$, define

$$E_\theta = \operatorname{diag}\left(\frac{1}{2}\frac{\partial_\theta s}{s}\right), \tag{48}$$

so that $D_\theta = E_\theta D$ and therefore

$$A_\theta = E_\theta G + G E_\theta, \qquad b_\theta = E_\theta b. \tag{49}$$

What Changes Computationally

The old lengthscale path uses

$$\frac{1}{2}\left(\operatorname{tr}(K^{-1}\partial_\ell K) - \alpha^*(\partial_\ell K)\alpha\right), \tag{50}$$

with a data-space Hutchinson estimator for the trace term and the quadratic term expressed through $\alpha = K^{-1}y$.

The proposed feature-space objective derivative instead uses

$$\frac{1}{2}\left[\operatorname{tr}(A^{-1}A_\ell) - \frac{1}{\sigma_n^2}\left(2\,\Re(b_\ell^*\beta) - \beta^* A_\ell \beta\right)\right]. \tag{51}$$

Only the trace part remains stochastic:

$$\operatorname{tr}(A^{-1}A_\ell) \approx \frac{1}{J}\sum_{j=1}^{J} v_j^* A^{-1} A_\ell v_j, \tag{52}$$

with feature-space Rademacher probes $v_j \in \{\pm 1\}^M$.

The right-hand side is now

$$A_\ell v = E_\ell G v + G(E_\ell v), \tag{53}$$

which requires only Toeplitz applications of $G = DCD$ and diagonal multiplications.

Why This Might Be Better

Relative to the older lengthscale trace estimator, this route has three potential advantages:

- it keeps the entire derivative in feature space;
- it avoids the large deterministic-constant minus noisy-correction structure that killed the previous trace-only rewrites;
- it replaces the data-space derivative probe path with Toeplitz-based applications of A_ℓ.

But there is also a real risk: A_ℓ is still generally sign-indefinite, so the Hutchinson trace estimator may still have high variance at very small J. That has to be checked empirically, not assumed.

Empirical Outcome

I tested this whole-objective feature-space estimator against exact ground truth for the approximate EFGP objective in the same style as the earlier lengthscale experiments:

- small real-data subsets with explicit dense K available;
- the frozen late-PRISM state using exact formed $M \times M$ systems.

The algebra checks out. On the small dense checks, the exact feature-space gradient matched the exact dense-K gradient to about 2×10^{-3} in absolute error.

But as an estimator, the picture is mixed:

- it is *much* better than the previous failed feature-space rewrites;
- it is still materially noisier than the old estimator at $J = 1$;
- its feature-space trace solve is harder for CG than the old lengthscale trace solve.

On the small SE check ($N = 192$, $M = 625$), with $J = 1$ over 1000 seeds:

- the old estimator had gradient mean/std approximately

$$3.268 \times 10^2 \ / \ 3.500 \times 10^2;$$

- the whole-objective feature-space estimator had gradient mean/std approximately

$$1.834 \times 10^2 \ / \ 4.611 \times 10^3.$$

On the small Matérn check ($N = 192$, $M = 3481$):

- the old estimator had gradient mean/std approximately

$$-1.987 \times 10^2 \ / \ 1.031 \times 10^2;$$

- the whole-objective feature-space estimator had gradient mean/std approximately

$$-1.444 \times 10^2 \ / \ 1.678 \times 10^3.$$

So on the small checks it is clearly worse than the current estimator.

On the frozen late-PRISM state ($N \approx 1.21 \times 10^7$, $M = 625$), with $J = 1$ over 20 seeds:

- the old estimator had gradient mean/std approximately

$$1.507167 \times 10^7 \ / \ 3.827 \times 10^3;$$

- the whole-objective feature-space estimator had gradient mean/std approximately

$$1.512696 \times 10^7 \ / \ 6.854 \times 10^5.$$

This is still far noisier than the old estimator, but it is no longer catastrophically bad. The relative standard deviation is about 4.5% on this PRISM state, versus about 2.5×10^{-4} for the old estimator.

The speed story goes the other way. On the same frozen PRISM state with the actual fast operators and Jacobi preconditioning:

- old lengthscale path: about 1.23s, with about 172 CG iterations;

- whole-objective feature-space path: about 0.17s, but about 1250 CG iterations, i.e. it hits the default $2M$ cap.

So the new estimator removes almost all of the N-scale work, but it replaces it with a substantially harder feature-space Krylov problem.

The practical conclusion is:

- this is *not* a clean drop-in replacement for the current lengthscale estimator;
- it is algebraically sound and dramatically cheaper in wall clock on very large datasets;
- but at low probe count it is still noisier than the old estimator, and its trace solve is harder for CG;
- if this direction is pursued further, it probably needs additional variance reduction or a stronger/recycled solver, not just the bare algebraic rewrite.